

Question 1

M2L9

In order to check if they are independent, we need to check every combination of x and y to see that $f(x, y) = f_X(x)f_Y(y)$. As soon as one of these is found that is not equivalent, independence can be ruled out.

Question 2

M2L11

Nothing to add to the steps given in the practice exam document.

Question 3

M4L3

Clock Movement

**Time flies like an arrow.
Fruit flies like a banana.**

Next-Event Time Advance: The clock is initialized at 0. All known future event times are determined and placed in the *future events list* (FEL), ordered by time.

The clock advances to the most *imminent event*, then to the next most imminent event, etc.

At each event, the system state and FEL are updated.

Question 4

M6L3 M6L6

Go through each combination of a and b values and check what the cycle lengths are.

Question 5

M6L3

If you have a calculator that performs the `mod` operation, then use that.

If you don't, then you could do it one of three ways:

Method 1

$$(16807 \times 1234321) \div (2^{31} - 1) \approx 9.66025193066$$

$$9.66025193066 - 9 \approx 0.66025$$

$$0.66025 \times (2^{31} - 1) = 1417880223.99$$

Round up to obtain 1417880224

Method 2

$$(16807 \times 1234321) = 20745233047$$

$$(16807 \times 1234321) \div (2^{31} - 1) \approx 9.66025193066$$

$$9 \times (2^{31} - 1) = 19327352823$$

$$20745233047 - 19327352823 = 1417880224$$

Method 3 (Use the operations from the FORTRAN implementation)

$$1234321 // 12773 = 9 \text{ (integer division)}$$

$$16807 \times (1234321 - 9 \times 127773) - 9 \times 2836 = 1417880224$$

Question 6

M6L4

Nothing to add to the steps given in the practice exam document.

Question 7

M6L4

Nothing to add to the steps given in the practice exam document.

Question 8

M6L3

Nothing to add to the steps given in the practice exam document.

Question 9

M6L9

Nothing to add to the steps given in the practice exam document.

Question 10

M6L8

The trivial completion of the table is using the fact that we know there are 1000 PRNs.

$$275 + 250 + 250 + ?? = 1000$$

$$?? = 1000 - (275 + 250 + 250) = 225$$

$$\chi_0^2 = \frac{(275 - 250)^2}{250} + \frac{(250 - 250)^2}{250} + \frac{(250 - 250)^2}{250} + \frac{(225 - 250)^2}{250}$$

$$= \frac{(25)^2}{250} + \frac{(-25)^2}{250} = 5$$

Question 11

M6L8

Percentage Points of the Chi-Square Distribution									
Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	10.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

Question 12

M2L5 M7L2 M7L3

$$U = (X - 1)^2$$

$$U^{1/2} = X - 1$$

$$X = U^{1/2} + 1$$

Question 13

M7L2 M7L3

To find the value of 0.6128, use the same process as described [here](#).

$\Phi^{-1}(0.27)$ is saying what value of z is it such that $P(Z \leq z) = 0.27$.

$$1 - 0.27 = 0.73$$

It's somewhere between 0.61 and 0.62

Standard Normal Table

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

$0.7324 - 0.7291 = 0.0033$ is the distance between these values.

$0.73 - 0.7291 = 0.0009$ is how far from 0.7291 that 0.73 is.

$0.0009/0.0033 = 0.27272727\dots$ gives us a proportion of how this relates to the distance between the two.

We apply this to the distance of $0.62 - 0.61 = 0.01$

$$0.01 \times 0.27272727\dots = 0.0027$$

$0.61 + 0.0027 = 0.6127$ which gets us close to the value from the solution of 0.6128

But since it's on the left hand side, make sure you make it negative.

Question 14

M7L2

It's not relevant what the distribution of X is. The inverse transform theorem states that $F(X) \sim \text{Unif}(0, 1)$

$$E[3F(X) - 1] = 3E[F(X)] - 1 = \frac{3}{2} - 1 = \frac{1}{2}$$

Question 15

M7L2 M7L4

Nothing to add to the steps given in the practice exam document.

Question 16

M7L12

Example: $Z_2/Z_1 \sim \text{Nor}(0, 1)/\text{Nor}(0, 1) \sim \text{Cauchy} \sim t(1)$.

Moreover,

$$Z_2/Z_1 = \frac{\sqrt{-2\ell\text{n}(U_1)} \sin(2\pi U_2)}{\sqrt{-2\ell\text{n}(U_1)} \cos(2\pi U_2)} = \tan(2\pi U_2).$$

Thus, we've just proven that

$$\tan(2\pi U) \sim \text{Cauchy} \quad (\text{and, similarly, } \cot(2\pi U) \sim \text{Cauchy}).$$

Question 17

M7L6

Review [Triangular Distribution Shifting and Scaling](#)

You should realize that this is a combination of two uniform random numbers which will result in a triangular distribution. However, since they are being subtracted, the method shown in the solution document utilizes the fact that U and $1 - U$ are both $\text{Unif}(0, 1)$ to change it from being a subtraction to an addition. The result is a $\text{Tria}(0, 1, 2)$ distribution shifted to the left by 1 unit.

Question 18

M7L6

Nothing to add to the steps given in the practice exam document.

Question 19

M7L2

If X has distribution $\Phi(x)$ this means that X is a standard Normal random variable. The CDF for the [standard Normal gets this special character](#) of Φ since it is so special and widely used.

We set $\Phi(X) = U$ and solve for X by taking the inverse of each side.

This results in $X = \Phi^{-1}(U)$ which means that X will have a standard Normal distribution.

Since the standard Normal distribution is centered at 0 and it is symmetric, then you should be able to immediately know that half the area is to the left of 0.

Using the table:

2

Standard Normal Table

1	Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
	0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
	0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
	0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
	0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
	0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
	0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
	0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
	0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
	0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
	0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
	1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
	1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
	1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
	1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
	1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
	1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
	1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
	1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
	1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
	1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
	2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
	2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
	2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
	2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
	2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
	2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
	2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
	2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
	2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
	2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

Question 20

M2L12 M7L2 M7L6

Recall that $-\frac{1}{\lambda} \ln(1 - U) \sim -\frac{1}{\lambda} \ln(U) \sim \text{Exp}(\lambda)$

Therefore, $-\ln(U_1) \sim \text{Exp}(1)$

Question 21

M7L2 M7L4

We're not concerned about the ≥ 4 case because the probability is so small.

Question 22

On some of the practice exams, there were questions related to pop culture, which were considered bonus questions. Please do not stress over these, as they are meant to be fun and easy free points. If questions such as these appear on actual exams, you will get credit for them regardless.

Question 23

M5L8

Decide Module



Decide

- When an entity gets to a Decide module, he can...
 - Randomly go to either of two locations (“2-way by Chance”) (probabilities are given in percentages)
 - Randomly go to any of various locations (“N-way by Chance”)
 - Go to either of two locations if a condition is satisfied (“2-way by Condition”)
 - Go to any of various locations if a condition is satisfied (“N-way by Condition”)

Question 24

M5L19

Eh??

- Yikes! Why are we seeing stuff like Seize, Delay, Release **yet again!?** At least two reasons...
 - Arena has been built in many layers over the years, and these are what's left from SIMAN – the “original” incarnation of Arena.
 - But in any case, and for whatever reason, certain primitive blocks such as the Queue block can't even connect to a Seize module from the Advanced Process template or a Process module from the Basic Process template. So you're stuck!
- **Demo Time!**
 - A Queue-Seize-Delay-Release example using primitive blocks (we'll see this again in the Call Center example coming up).
 - Using the Alter block to change the number of resources.



Question 25

M5L15

The Story (cont'd)

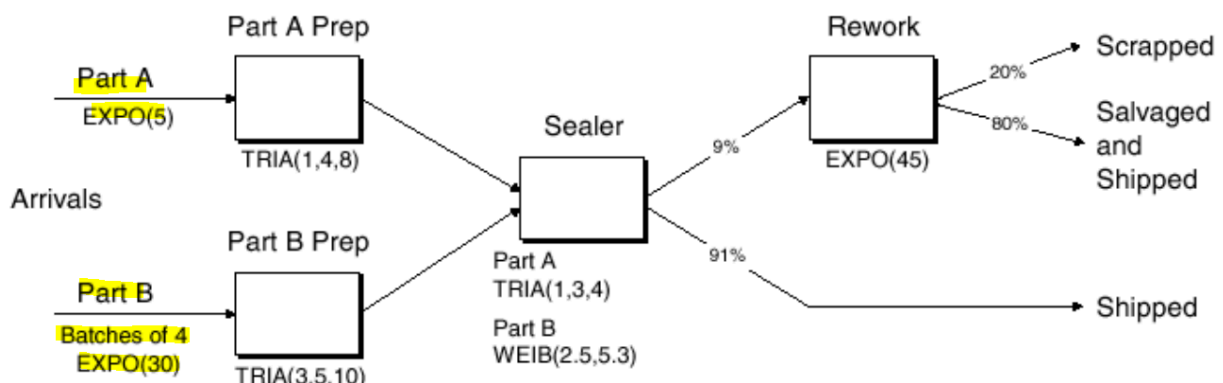
- How to handle different A and B service times at Sealer?
 - Trick 1: **Pre-assign** the service times as an **attribute** (Sealer Time) in an Assign module immediately after each customer arrives. Use that attribute regardless of being a Type A or B part.
 - Trick 2: While we're at it, use the Assign to store each customer's arrival time as an **attribute**. Use the Arena variable TNOW to do so.
- Record departure times just before parts get Dispose'd, This will allow us to get average cycle times (depart – arrival times) for any of the 3 types of parts (pass on first try, pass on second, fail both).

Question 26

M5L15

The Story (cont'd)

I stole this flowchart from KSZ...



Question 27

M5L16

Fake Customers

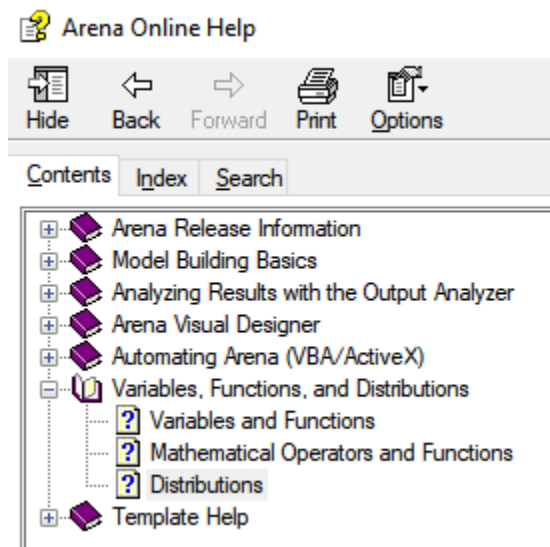
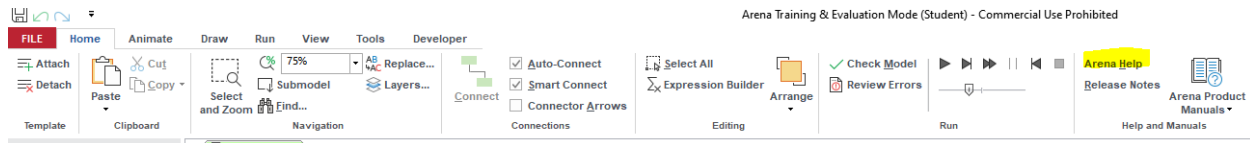
- You can use “fake” customers to accomplish various tasks during a simulation.
- Not actual customers that you care about in terms of waiting times or use of resources.
- **Demo Time** will explain all!
 - Calculate normal probabilities
 - Keep track of which time period you’re in (part of Call Center example coming up)
 - Breakdown demon

Question 28

M2L7 M2L12 M5L23

The documentation for DISC is instructive on how to use this function.

You can either search for it, or go to Variables, Functions, and Distributions -> Distributions.



Empirical DISCRETE Distribution

DISCcrete(Prob1,Value1,Prob2,Value2, . . . [,Stream])

Prob_i — Cumulative probability

Value_i — Value associated with probability Prob_i

Parameter Set — Parameter set ID containing values for Prob_i and Value_i pairs

When evaluating the DISCcrete function, [SIMAN](#) generates a random number between 0 and 1. SIMAN then searches the probabilities, Prob_i, until it finds one larger than the random number. The Value_i associated with that probability is returned.

Example: DISC(0.3,1, 0.8,2, 1.0,3) This function returns the value 1 approximately 30 percent of the time, the value 2 approximately 50 percent of the time (0.8 - 0.3 = 0.5 or 50 percent), and the value 3 approximately 20 percent of the time (1.0 - 0.8 = 0.2 or 20 percent).

Drawing a random value from the empirical discrete distribution uses a sequential search through the cumulative probabilities table. Let (P_k, V_k) be a (cumulative probability, value) pair. The search algorithm proceeds as follows:

```
Initialize k=1
Generate U ~ U( 0,1 )
if U < Pk, then
    DISCRETE = Vk
    and return
Otherwise, increment k and repeat check.
```

Note that the value returned is the lowest V_k value associated with the first cumulative probability, P_k, that is greater than the sample drawn from the uniform distribution. This contrasts with the empirical continuous distribution which interpolates between probability values to obtain a random value.

For the function in this question, we have DISC(0.5, 2, 1, 3). This corresponds to the following CDF:

$$F(x) = \begin{cases} 0 & x < 2 \\ 0.5 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Using the example in the documentation, we can deduce that this will return the value 2 50% of the time and the value 3 (1 - 0.5 = 0.5) 50% of the time.

And the corresponding PDF:

$$f(x) = \begin{cases} 0.5 & x = 2 \\ 0.5 & x = 3 \\ 0 & \text{otherwise} \end{cases}$$

The expected value of this random variable is $E[X] = \sum_x x f(x) = (2)(0.5) + (3)(0.5) = 2.5$

In Arena, the EXPO function is parameterized by the mean, so the answer is 2.5.

Exponential Distribution

EXPONential(Mean[,Stream])

Mean — Mean or scale parameter beta

Parameter Set — Parameter set ID containing the value of Mean

The (literal)inverse-transform algorithm implemented is:

Generate $U \sim U(0,1)$

Set EXPONENTIAL = $-\beta * \ln(1 - U)$

Question 29

M5L11

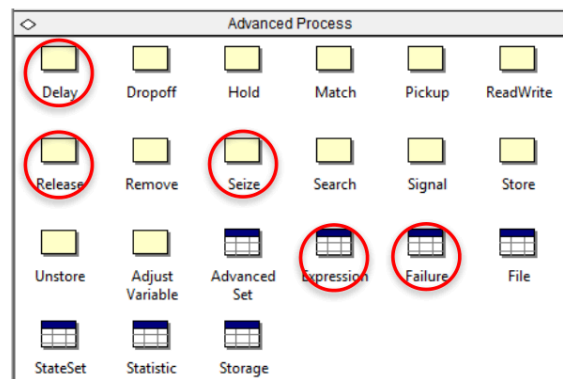
Nothing to add to the steps given in the practice exam document.

Question 30

M5L17 M5L19

Advanced Process Template

- File > Template Panel > Attach (and hopefully you find it in the Rockwell Software \ Arena \ Template directory).
- Lots of new modules and spreadsheets.
- I don't expect you to memorize them all.
- Concentrate for now on a few of these (comment on others later):
 - Seize, Delay, Release modules
 - Expression, Failure spreadsheets.



Blocks Template

- File > Template Panel > Attach (and hopefully you find it in the Rockwell Software \ Arena \ Template directory).

- Huge number of blocks!
- Do not be scared!
- They're sort of a self-contained language (related to SIMAN).
- Very specialized and low-level.
- Very useful, but we'll only need a few for now:
 - Seize, Delay, Release, Queue, Alter

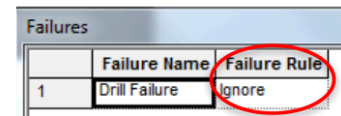
Blocks			
☐ Access	☐ Activate	☐ Allocate	☐ Alter
☐ Assign	☐ Begin	☐ Block	☐ Branch
☐ Capture	☐ Close	☐ Combine	☐ Convey
☐ Copy	☐ Count	☐ Create	☐ Delay
☐ Detect	☐ Dispose	☐ Dropoff	☐ Duplicate
☐ Else	☐ Elself	☐ EndIf	☐ EndWhile
☐ Event	☐ Exit	☐ FindJ	☐ Free
☐ Group	☐ Halt	☐ If	☐ Include
☐ Insert	☐ Match	☐ Move	☐ OnChange
☐ PickQ	☐ Pickup	☐ Preempt	☐ Proceed
☐ QPick	☐ Queue	☐ Read	☐ Release
☐ Relinquish	☐ Remove	☐ Request	☐ Route
☐ Scan	☐ Search	☐ Seize	☐ Select
☐ Signal	☐ Split	☐ Stack	☐ Start
☐ Station	☐ Stop	☐ Store	☐ Tally
☐ Trace	☐ Transport	☐ Unblock	☐ Unstore
☐ VBA	☐ Wait	☐ While	☐ Write
☐ Zap			

Question 31

M5L18

Failures (cont'd)

- Go to Resource spreadsheet in the Basic Process template.
- Click on Failures column.
- Add a Failure Name.
- Choose Failure Rule (will discuss in a minute).
- Go to the Failure spreadsheet in the Advanced Process template, where you'll see your new failure name.
- Choose type of failure:
 - Count (failure after a certain # of arrivals)
 - Time (after a certain amount of time)
- Choose downtime (for repair): Can be any expression.



Failures		
	Failure Name	Failure Rule
1	Drill Failure	Ignore

Question 32

M5L18

Remarks

- Can schedule multiple failures by using multiple rows of the Failures column in the Resource spreadsheet, e.g., type I failure, type II failure, scheduled maintenance.
- Types of Failure Rules:
 - Ignore (complete service of current customer, but **reduce** repair time). E.g., if repair time = 1 hour and cust still needs 10 min, then repair time gets reduced to 50 min and finishes at 60 min mark.
 - Wait (complete service of current cust and **delay** repair). E.g., if repair time = 1 hr and cust needs 10 min, then repair finishes at 70 min mark.
 - Preempt (stop service of current cust, but **complete service after the repair**). Repair stops at 60 min mark, cust finishes at 70 min mark.

Question 33

M5L20

Resource Sets

- Use the Set spreadsheet in the Basic Process template to define sets.
- We'll just study resource sets for now.
- A “vanilla” **resource** has identical, interchangeable servers. But a **resource set** can have **distinct** servers, with different schedules, service speeds, service specialties, etc.
- Call Center Example: Three products with the following resources
 - Product 1: Charity, Noah, Molly, Anna, Sammy
 - Product 2: Tierney, Sean, Emma, Anna, Sammy
 - Product 3: Shelley, Jenny, Christie, Molly, Anna, Sammy
 - Molly, Anna, and Sammy all have degree of cross-functionality.
 - All 11 servers turn out to have different schedules (more on that later)



Question 34

M5L21 M5L22

Call Center Description

Program is arranged in [submodels](#).



- Create and Direct Arrivals
 - How often do calls show up and where do they go?
- Tech Support Calls
 - What kind of tech support do you need?
- Returned Tech Calls
 - Sometimes the guy has to get back to you.
- Sales Calls
- Order Status Calls
- Time Period Counter – What ½-hour period of the day is it?

Question 35

M5L21 M5L22

Arrivals

Calls show up according to a **nonhomogeneous Poisson process**.

- The center accepts calls from 8:00A till 6:00P.
 - The arrival rates change every half hour (see table below from KSZ).
 - Also see **Arrival Schedule** in the Basic Process Schedule spreadsheet.

Time	Rate	Time	Rate	Time	Rate	Time	Rate
8:00 – 8:30	20	10:30 – 11:00	75	1:00 – 1:30	110	3:30 – 4:00	90
8:30 – 9:00	35	11:00 – 11:30	75	1:30 – 2:00	95	4:00 – 4:30	70
9:00 – 9:30	45	11:30 – 12:00	90	2:00 – 2:30	105	4:30 – 5:00	65
9:30 – 10:00	50	12:00 – 12:30	95	2:30 – 3:00	90	5:00 – 5:30	45
10:00 – 10:30	70	12:30 – 1:00	105	3:00 – 3:30	85	5:30 – 6:00	30



← These rates are **per hour**
(even though time intervals
are 30 min)

- A few staff actually stay at work until 7:00P, in order to let the calls at the end of the day clear out. (You have to explicitly model 0 arrivals for the last two half-hour segments of the Arrival Schedule.)



Question 36

M5L21 M5L22

Tech Support Info

Tech Support staff have some interesting issues.

- They all have 8 hr days + 30 min for lunch.
 - They all have different schedules. We'll need Resource Schedules to model this on an individual basis.
 - They all have different product expertise. We'll need Sets to model this. Here are the Preferred Orders for the sets:
 - Product 1: Charity, Noah, Molly, Anna, Sammy
 - Product 2: Tierney, Sean, Emma, Anna, Sammy
 - Product 3: Shelley, Jenny, Christie, Molly, Anna, Sammy
- 

Question 37

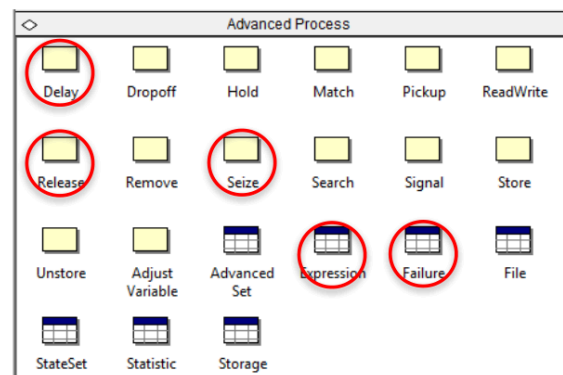
M5L9

Question 38

M5L17

Advanced Process Template

- File > Template Panel > Attach (and hopefully you find it in the Rockwell Software \ Arena \ Template directory).
- Lots of new modules and spreadsheets.
- I don't expect you to memorize them all.
- Concentrate for now on a few of these (comment on others later):
 - Seize, Delay, Release modules
 - Expression, Failure spreadsheets.



Question 39

M5L23

Description of (s,S) Policy

- Simulate the widget inventory stock over time.
- Use modules from the Basic and Advanced panels.
- Customer interarrivals are $\text{EXPO}(0.1)$.
- Demand size is a DISC function.
- Demand is always “met” (backlogged, if necessary).
- Inventory is “taken” at the start of each day.
- If inventory is below s , we order up to S .
- Delivery lead time is $\text{UNIF}(0.5,1)$ before order arrives (pretty quick, but by the time it arrives, other customers will have arrived).
- Order costs (set-up + incremental), holding costs, penalty costs.